

Quiz 0 – a sample quiz

Instructions: You must show all of your work to receive credits for correct answers.**Problem 1.** (4 points) Find the following limits using the **L'Hospital's Rule**:

(a) $\lim_{x \rightarrow +\infty} \frac{\ln x}{\sqrt{x}}$

$$= \lim_{x \rightarrow +\infty} \frac{(\ln x)'}{(\sqrt{x})'} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{(\frac{1}{2}\frac{1}{\sqrt{x}})}$$

$$= \lim_{x \rightarrow +\infty} \frac{2}{\sqrt{x}} = 0$$

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{\sin x}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{(\sin x)'}{(2x)'} = \lim_{x \rightarrow 0} \frac{\cos x}{2}$$

$$= \frac{1}{2} \quad (\text{use L'Hospital's Rule twice!})$$

Problem 2. (6 points) Find the derivatives of the following functions:

(a) $y(x) = \sqrt[3]{x} + \ln x$

$$y'(x) = (x^{\frac{1}{3}})' + (\ln x)'$$

$$= \frac{1}{3} x^{\frac{1}{3}-1} + \frac{1}{x}$$

$$= \frac{1}{3} x^{-\frac{2}{3}} + \frac{1}{x}$$

(b) $f(t) = t^2 \sin t$

$$f'(t) = (t^2)' \sin t + t^2 (\sin t)'$$

$$= 2t \sin t + t^2 \cos t$$

(Product Rule)

(c) $h(x) = e^{x^2+1}$

$$h'(x) = e^{x^2+1} \frac{d}{dx} [x^2+1]$$

$$= 2x e^{x^2+1}$$

(Chain Rule)

(d) $g(x) = \left(\frac{x+1}{x-1} \right)^6$

$$g'(x) = 6 \left(\frac{x+1}{x-1} \right)^5 \frac{d}{dx} \left[\frac{x+1}{x-1} \right]$$

$$= 6 \left(\frac{x+1}{x-1} \right)^5 \frac{(x+1)'(x-1) - (x+1)(x-1)'}{(x-1)^2}$$

$$= 6 \left(\frac{x+1}{x-1} \right)^5 \frac{(x-1) - (x+1)}{(x-1)^2}$$

$$= 6 \left(\frac{x+1}{x-1} \right)^5 \left(-\frac{2}{(x-1)^2} \right) = -12 \frac{(x+1)^5}{(x-1)^7}$$

(Chain Rule + Quotient Rule)